PURE PATH LAB | LABORATORY MANAGEMENT SYSTEM

Software Requirements Specification

Single-laboratory pathology and basic radiology system
Product scope, functional requirements and acceptance criteria

Version 1.0 • 27 October 2026 • Issued for technical review and acceptance

SYSTEM SCOPE	USER ACCESS	OPERATING MODEL
Pathology and basic USG / X-ray	Staff accounts and patient OTP portal	Responsive, online web application
Registration, billing, reports and QR verification	Role-based staff permissions	Browser access and PDF / print outputs

Specification purpose

This document defines the Pure Path Lab system requirements, functional scope, technical dependencies and acceptance checks for the laboratory management system.

The requirements below provide the technical scope and a shared basis for implementation and acceptance of the Pure Path Lab laboratory management system.

DOCUMENT CONTROL	DETAIL
System	Pure Path Lab Laboratory Management System
Scope	Single-laboratory pathology and basic radiology (USG / X-ray)
Technology baseline	React 18 / Vite • Express 4 • Mongoose 8 • MongoDB Atlas
Deployment baseline	Browser application with hosted API and database
Version and status	1.0 • 27 October 2026 • Issued for technical review and acceptance

Document guide

How to read the system scope and requirements.

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Document conventions

- **SHALL** identifies a binding requirement for the delivered system.
- **FR** means functional requirement; **NFR** means non-functional requirement; **AC** means acceptance check.
- Items explicitly identified as minimal or as-is are retained at their stated current scope.
- Acceptance checks refer to the applicable FR and NFR identifiers in this document.

1. Introduction

Purpose, scope and intended use of this specification.

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and quality requirements, system context, dependencies and acceptance checks for the Pure Path Lab Laboratory Management System (LMS). The system supports a single pathology laboratory with basic USG and X-ray workflows.

1.2 Product scope

The system covers staff access, patient registration, billing and collections, test masters, report entry and sign-off, report printing and PDF generation, QR verification, barcodes and sample labels, business accounts, patient OTP access, notifications, and basic USG / X-ray case handling.

1.3 Scope boundary

This SRS defines the product requirements and delivery acceptance baseline. Existing supporting features explicitly labelled minimal or as-is will be retained with limited functionality. Requirements outside this scope require an update to this specification before implementation.

1.4 Intended readers

READER	USE
Laboratory owner / client	Review the system scope, dependencies, and acceptance checks.
Delivery team	Implement and verify the requirements in this SRS.
UAT reviewers	Run the acceptance checks against the delivered system and record results.

1.5 Definitions

TERM	MEANING
Case / bill	A billing event linked to one or more laboratory reports.
TAT	Turnaround time, expressed in hours.
Verify / sign	Pathologist sign-off that finalises a report.
UAT	User acceptance testing performed by the client.
DLT / Sender ID	Indian SMS delivery registration and approved sender details required for live SMS.
WABA	WhatsApp Business Account; WhatsApp messaging is excluded from this phase.

2. Product context

The system boundary, operating environment and user responsibilities.

2.1 Product overview

Pure Path Lab is a browser-based, online application. A React single-page interface communicates with an Express REST API and MongoDB database. The system generates laboratory PDFs and QR verification data, and accepts report images and signatory images through upload workflows.

2.2 User roles

ROLE	PERMITTED WORK
Administrator / superadmin	Application settings, laboratory profile, employees and permissions, rates, accounts, and report verification.
Pathologist	Review, verify, sign, reject, or request changes to reports; manage assigned signatures.
Technician / laboratory staff	Register patients, prepare bills, collect samples, enter results, upload images, and print.
Radiologist / USG / X-ray staff	Work modality cases, upload images, enter findings, preview reports, and sign off.
Accountant / cashier	Manage payments, dues, voids, cashbook, and expenses.
Patient	Use mobile number and OTP to view or download their own reports and bills. Patients do not have staff accounts.

2.3 Operating environment

- Desktop-first responsive web application; patient report access shall be mobile-friendly.
- Online use. Staff printing targets A4 laser or inkjet printers; labels use the approved sticker stock.
- Supported browsers: current Chrome, Edge, and Firefox.

2.4 Technology baseline

LAYER	BASELINE
Frontend	React 18, React Router v6, Axios, and Vite.
Backend	Node.js and Express 4 REST API, JWT staff authentication, file upload, PDF generation and QR generation.
Data	MongoDB via Mongoose 8; Atlas Free M0 at launch.
Hosting	Vercel static frontend and API rewrites.
Patient authentication	Mobile number and OTP session; MSG91 for live OTP delivery, subject to valid credentials and DLT readiness.
Email	SMTP account or email provider integration.

3. Functional requirements

Staff identity, patient records, and laboratory test configuration.

FR-01 Staff authentication and employee management

- The application SHALL support staff login by email and password and issue authenticated staff sessions using JWT.
- The application SHALL provide employee creation, viewing, updating, activation and deactivation, role and department assignment, designation and joining details, and password reset.
- The application SHALL enforce access permissions on the server. The role and permission interface SHALL reflect the enforced access rules.
- Session listing and revocation SHALL be completed within the build.
- Rate limits SHALL be applied to authentication attempts. OAuth integration hooks may be configured where provider credentials are supplied.

FR-02 Patient management

- Staff SHALL be able to register patient demographic and contact information and link a referring doctor or agent.
- Staff SHALL be able to search patient records and view a patient’s related case, bill, and report history.

FR-03 Laboratory test master

- Authorised users SHALL manage test code, name, department, rate, unit, method, normal ranges, critical limits, age or sex applicability, turnaround time, and interpretation text.
- The test master SHALL support derived-test indicators and formulas, plus categories, panels, and packages.
- The build SHALL provide bulk rate revision. The source implementation currently supports individual rate changes; bulk revision is included in the system scope.

> **Access and patient privacy**
>
> Patient access is restricted to the reports and bills returned for that patient’s verified mobile number. Staff permissions are checked by the backend and SHALL not rely solely on hiding interface controls.

3. Functional requirements

Billing, business records, referral tracking and laboratory settings.

FR-04 Billing, payments and case creation

- Staff SHALL create a bill from searchable tests, panels or packages, with applicable rates, discounts, invoice numbering and paid, partially paid, pending or due status.
- Cashiers SHALL record full or partial collections and payment modes, review dues, and void a bill with a recorded reason.
- The application SHALL print invoices and generate a bill barcode. Each new bill SHALL link to the corresponding case and report workflow.
- The cashbook SHALL support filters and date-range summaries. Staff SHALL be able to record and review laboratory expenses.

FR-05 Accounts and business reporting

- Authorised staff SHALL review account balances and ledger transactions, outstanding dues, and department-filtered business records.
- The application SHALL provide referral and agent tracking, a daily business view, and test-usage analytics.
- Dataset export is included. JSON export is present in the source build; CSV export and export history are to be completed.

FR-06 Doctors, agents and laboratory settings

- Authorised staff SHALL manage referring doctors and link referrals to cases; agent and collector tracking SHALL be supported.
- Administrators SHALL configure laboratory identity, address, contact information, registration details, branding, and report numbering (default prefix PPL-).
- SMS notification templates SHALL be configurable through the settings interface.

Billing integrity

A payment entry shall be associated with the relevant bill and reflected in its paid, partial, or due balance. Void actions require a reason. The cashbook summary shall reconcile to recorded transactions for the selected test date range (acceptance check AC-03).

3. Functional requirements

Pathology reporting, verification, signatures and turnaround time.

FR-07 Report lifecycle and results

- The system SHALL generate linked reports from bills and provide Today, Due and Pending worklists with search and department filtering.
- Worklists SHALL display turnaround-time countdowns and due-soon or overdue indicators.
- Staff SHALL enter test results. Results outside configured ranges SHALL display HIGH, LOW, or CRITICAL flags as applicable.
- Pathologists SHALL be able to sign and verify reports, reject a report with a reason, or request changes with comments. Unsigned reports and a verification screen SHALL be available.
- The report sign-off action SHALL safely handle repeat submission, such as a double click.

FR-08 Report output and public verification

- Report preview and final PDF or print SHALL use the same output template.
- Final reports SHALL display configured laboratory letterhead, required report details, and applicable signatory information.
- The final report SHALL include a QR code containing a public verification URL. Public verification SHALL use a token URL and SHALL NOT expose internal database identifiers.
- Barcode labels, case and sample labels, and a sticker sheet SHALL be completed for the stock size agreed before UAT.

FR-09 Signatures and turnaround-time configuration

- Administrators SHALL create and assign signatory signatures by department. Reports SHALL use the selected signature at sign-off and render it in the output.
- Turnaround time SHALL be configurable per test and report, with default and urgent or emergency settings.
- Due and overdue status SHALL be calculated consistently using the laboratory’s local time convention.

> **Output baseline**
>
> The client supplies laboratory identity, registration details, disclaimer, logo, and signatory image files before the final template and UAT round. A4 is the report target; sticker layout depends on approval of the actual label stock.

3. Functional requirements

Radiology cases and retained supporting workflows.

FR-10 USG and X-ray case handling

- Authorised staff SHALL create and manage USG and X-ray cases and assigned worklists.
- Staff SHALL enter findings, upload relevant images with upload progress and retry feedback, preview the report output, and record sign-off.
- The system SHALL support image deletion and inclusion of images within report PDFs.

FR-11 Doctor portal, appointments and support

- Existing basic doctor portal, appointment or booking flow, and support ticketing SHALL be retained in their current minimal working state.
- An advanced doctor portal and slot-availability engine are not part of the scope.

FR-12 Audit trail and notification jobs

- The application SHALL provide an activity or audit-log viewer once the planned event capture is implemented.
- The application SHALL provide visibility into failed notification jobs and retry status once the planned job handling is implemented.

3.1 Workflow summary

WORKFLOW	EXPECTED SEQUENCE
Pathology	Patient → bill and sample → enter results → review and sign → PDF / print → QR verification.
USG / X-ray	Create case → upload images → enter findings → preview → sign off.
Patient access	Verified mobile number → OTP → own report and bill list → view or download.
Notifications	Select patient / doctor recipients → send using SMS or email template → record delivery outcome.

Retained scope

Minimal or as-is features are included only as described in this section. Additional functionality requested for a retained feature is documented through the requirements update process in Section 7.3.

3. Functional requirements

Patient OTP access, notifications and external interfaces.

FR-13 Patient OTP portal and public verification

- Patients SHALL sign in by mobile number and a one-time password; patient password accounts are not required.
- An OTP SHALL expire after five minutes by default. Verification SHALL allow at most three attempts by default, and resend requests SHALL be rate-limited.
- A successfully verified patient SHALL be able to list, view and download their own reports and bills.
- A final report QR scan SHALL open token-based public verification for that report.
- Development and testing may use provider test credentials while live MSG91 credentials and DLT approval are pending.

FR-14 SMS, email and delivery status

- The system SHALL support template-driven SMS and email sends from the report workflow, for one or multiple recipients, including patient and doctor recipients and optional cc / bcc.
- The system SHALL show Sent, Failed or Retry status and provide per-report delivery history.
- A failed notification SHALL NOT be shown as successfully delivered.
- The build SHALL connect MSG91 and the approved sender details once the client supplies credentials and DLT approvals.
- WhatsApp messaging and WABA configuration are excluded from this phase.

3.2 External service interfaces

SERVICE	SYSTEM INTERFACE
MSG91	OTP and SMS delivery; valid credentials and approved DLT sender details are required for live use.
Email	SMTP or provider API for report and workflow email.
Vercel	Frontend hosting and API routing target.
MongoDB Atlas	Hosted database environment.
Domain	Production URL and HTTPS access.
WhatsApp	Excluded from this system scope.

4. Non-functional requirements

Baseline expectations for usability, security and output quality.

ID	REQUIREMENT
NFR-01	Responsive interface; agreed screens shall work at 320 px mobile, tablet, and desktop widths without horizontal overflow; touch controls remain usable.
NFR-02	API access is authenticated. Staff permissions are enforced server-side. Passwords are not exposed. Authentication and OTP endpoints are rate-limited.
NFR-03	Report preview and final print / PDF use a single output layout. QR verification data remains readable in the PDF.
NFR-04	Repeated report-sign requests are safe. A failed notification is never represented as a successful send.
NFR-05	Report due and TAT calculations follow one configured laboratory local-time convention.
NFR-06	Public-facing pages include basic search metadata and a Lighthouse hygiene snapshot.
NFR-07	The system supports current Chrome, Edge and Firefox and A4 printing on laser or inkjet printers.
NFR-08	The build is deployed to the configured hosting and database environment after required access is available.

4.1 Security and access principles

Role-based permissions are checked by backend middleware for protected operations. OTP access is rate-limited, expires, and limits verification attempts. Public verification uses an opaque token URL. Client account ownership and access credentials are managed by the client.

4.2 Operational limitations at launch

The launch configuration uses Vercel and MongoDB Atlas Free (M0). Free-tier storage, backup, memory, and usage limits apply at launch. File-storage persistence depends on the selected hosting configuration; the deployment must be reviewed against operational storage and backup needs.

4.3 Printing and report output

A4 is the standard report page. Barcode and QR content must remain legible when printed. Sticker dimensions are configured after the client approves the target stock. USG and X-ray previews shall match their printed or downloaded output. The client supplies accurate test ranges and laboratory review/signature content.

4.4 External service boundaries

Availability, delivery times and service limits of Vercel, MongoDB Atlas, MSG91, the domain registrar and any email provider are governed by those providers. Formal security audits, penetration tests and compliance certifications are outside the technical scope of this SRS.

5. Delivery, dependencies and risks

Client-provided inputs and environment setup needed for go-live.

TECHNICAL INPUT / DEPENDENCY	SYSTEM AREA
Approved laboratory name, address, phone numbers, registration number, timings, disclaimer and logo	Letterhead, reports, invoices and public verification page.
Test masters, rates, doctors and agents	Configuration and master-data setup.
Labsmart exports and sample report PDFs	Master import and report-template alignment. Full history migration is outside this SRS.
MSG91 integration credentials and approved DLT Sender ID / template	Live OTP and SMS delivery; test credentials may be used during development.
Staff list, pathologist names / titles and signature images	User setup, permissions and report sign-off. Initial configuration covers up to 10 logins and 1–2 signatories.
SMTP credentials or email provider API	Email notification delivery.
A4 printer and sticker stock dimensions	Report and label layout validation.
Access to the selected deployment, database, domain and external service environments	Integration and production deployment.
One authorised contact to consolidate system decisions	Clarification of requirements and UAT results.

5.1 Data migration boundary

System setup includes one import of agreed master data (tests, rates, and doctors) from supplied exports. Full historical patient, bill, and report migration is outside this SRS. Additional migration requirements can be documented separately after review of the available data exports.

5.2 Technical risks and responses

RISK	RESPONSE
DLT / Sender ID approval delay	Use MSG91 test credentials during development; enable live OTP when approved sender details are available.
Export format or history limits	Import agreed masters first; review actual CSV data to define any historical migration.
Hosting or database tier limits	Review storage, backup and usage requirements against the selected platform configuration.
Missing laboratory content or signatories	Use supplied approved interim content during setup; confirm report content and signatories before UAT.

6. Acceptance checks

The client and delivery team use these observable checks during UAT.

ID	ACCEPTANCE CHECK	REQUIREMENTS
AC-01	Staff login → register a patient → create a bill → enter results → sign → produce report PDF / print with letterhead and QR.	FR-01 to FR-03, FR-07 to FR-09
AC-02	Patient verifies a mobile number by OTP and views / downloads their report when live MSG91 setup is available.	FR-13
AC-03	Record full, partial and due payments; void a bill with a reason; reconcile cashbook totals for a selected test date range.	FR-04, FR-05
AC-04	Create USG and X-ray cases, upload an image, preview and complete sign-off.	FR-10
AC-05	Print a bill barcode and verify a report by scanning its QR; check labels on approved sticker stock.	FR-07 to FR-08
AC-06	Deploy the application to the agreed production environment, connect the domain, and confirm administrator access.	FR-01 to FR-14

6.1 Acceptance procedure

For each acceptance check, reviewers record pass, fail or blocked status and the related requirement ID. A failure should include the affected workflow and steps to reproduce it. Acceptance checks verify the system requirements.

6.2 Defect classification

A defect is delivered behaviour that fails to meet an explicit requirement or acceptance check in this SRS. A request to add behaviour not stated in this SRS is a scope change and should be documented in an updated requirements baseline before implementation.

6.3 Traceability

REQUIREMENT GROUP	VERIFIED BY
FR-01 to FR-03: access, patients, test master	AC-01, AC-06
FR-04 to FR-06: billing, business, settings	AC-01, AC-03, AC-06
FR-07 to FR-09: reports, signatures, TAT	AC-01, AC-05
FR-10: USG / X-ray	AC-04
FR-11 to FR-12: retained support functions	AC-06
FR-13: patient portal and OTP	AC-02
FR-14: notifications	AC-01, AC-02; provider credentials required

7. Scope boundaries and dependencies

In-scope workflows, technical boundaries and external system interfaces.

7.1 External service interfaces

SERVICE	TECHNICAL ROLE
MSG91 / DLT	OTP and SMS delivery. Live integration requires valid credentials and approved sender details.
Email provider	Report and workflow email delivery by SMTP or provider API.
Vercel	Application hosting and API routing target.
MongoDB Atlas	Application database environment.
Domain / HTTPS	Production URL resolution and encrypted browser access.
WhatsApp / WABA	Excluded from the system scope.

7.2 Out of scope

- Multi-branch or SaaS tenancy, subscriptions, trials and paywalls.
- Inventory, stock control, HR, payroll, or instrument interfacing such as HL7 / ASTM.
- Native Android or iOS applications or a phlebotomist mobile application.
- Advanced analytics or business intelligence and payment-gateway collection, including Razorpay.
- Public marketing or landing pages; public-facing copywriting, photography and logo creation.
- Regional-language report templates; report language in this phase is English.
- WhatsApp / WABA messaging, formal security audit, penetration testing and compliance certification.
- Full historical patient, bill and report migration; additional migration requirements require a separate data specification.
- Any functionality not stated in this SRS.

7.3 Requirement changes

When a requirement changes or new behaviour is requested, record the change in an updated requirements baseline. The affected functional requirement and acceptance check should be updated before that change is implemented.

Appendix A. Implementation reference

Informational inventories from the implementation baseline.

These inventories identify the data areas and API groups supporting the requirements. They describe the implementation baseline and do not create functionality beyond Section 3.

A.1 Data entities

Activity; Agent; Bill; BillItem; Branch (minimal); DeliveryAttempt; Doctor; DoctorInvite (minimal); Expense; Inquiry (minimal); Interpretation; Job; LabProfile; ModalityCase; NotificationTemplate; OnboardingProgress (minimal); OtpSession; Patient; Report; ReviewRequest (minimal); Signature; SmsCredit; Subscription (scaffolding only); SupportTicket; TatConfig; Test; TestCategory; TestPackage; TestPanel; Transaction; User; USGCase; WebBrowser (minimal); XrayCase.

A.2 API route groups

Authentication; employees and roles; patients; bills and cashbook; reports; tests / categories / panels / packages; USG; X-ray; modality cases; doctors; doctor portal / appointments; transactions; expenses; dashboard; analytics; exports; notifications; public verification; patient portal; settings; laboratory setup; support tickets; background jobs; audit log; minimal branches; enquiries.

Appendix B. Technical review and acceptance

Review confirms that the system requirements, scope boundaries, and acceptance checks in this SRS have been understood.

REVIEW ROLE	NAME	SIGNATURE	DATE
Client technical reviewer			
Delivery technical reviewer			

Review notes

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